AUTHORIZATION AND PAYMENT INFRASTRUCTURE FOR AGENTIC SYSTEMS

Kaldrix

The approval desk for the age of AI agents.

Every action checked. Every payment and every record protected. Every decision remembered.

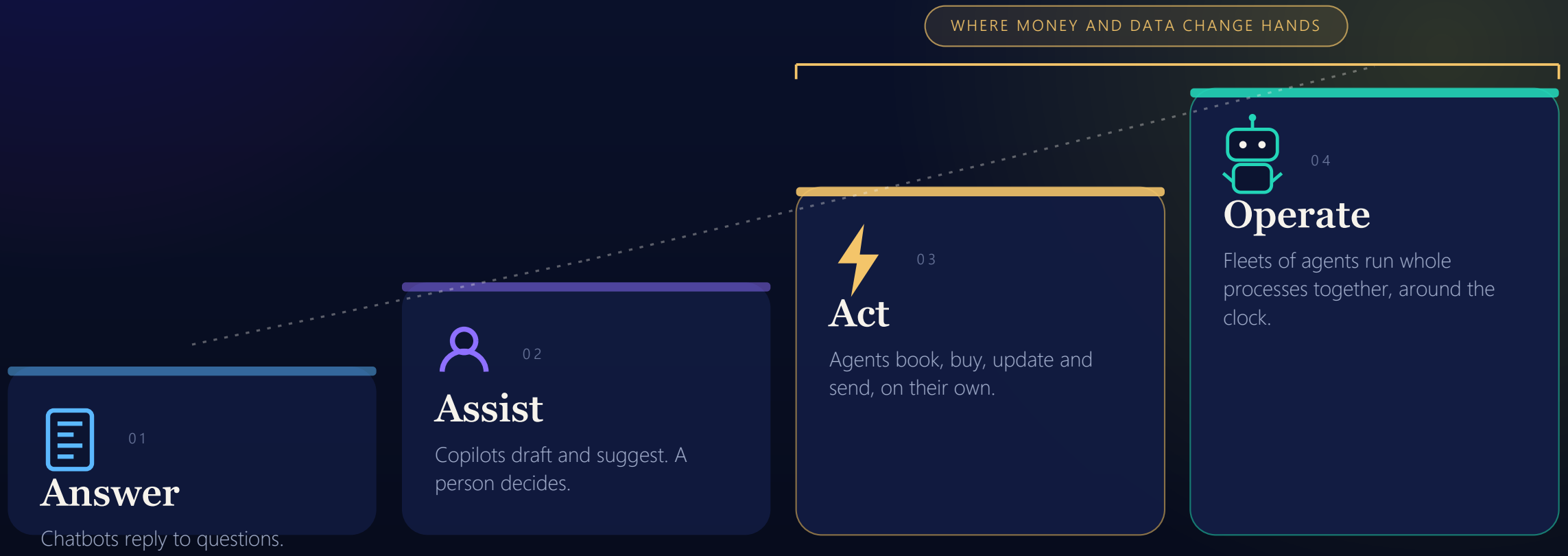
KALDRIX DECIDES AND GUARDS THE DATA. ZENTRIX MOVES THE MONEY.

CHECKED PROTECTED REMEMBERED



THE SHIFT

AI has stopped answering. It now acts.



THE PROBLEM IN ONE PICTURE

Two helpers. One account. Nobody in charge.

I will spend $40,000.



AGENT A

Looked at $50,000 a moment ago.



COMPANY BUDGET

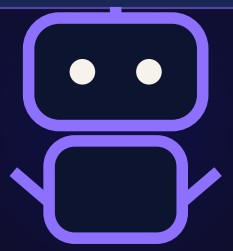
$50,000

$50,000 – $40,000 – $40,000

–$30,000

OVERSPENT

Me too: $40,000.



AGENT B

Looked at $50,000 a moment ago.

IN PLAIN WORDS

Each helper made a sensible choice from what it saw. Together they spent money that was never there.

UNDER THE HOOD

A stale read plus a non-atomic budget check: the classic check-then-act race, now with an agent holding real credentials.

THE ANSWER

Put one desk in the middle.



AGENT A · APPROVED

$40,000 spent

$10,000 left in the budget.

AGENT B · REFUSED

Stopped, with a reason

The budget changed after B looked. B is told why, and can look again.

IN PLAIN WORDS

Nothing is overspent, and the second helper is told exactly why it was stopped.

UNDER THE HOOD

Authorization is evaluated against the version the agent read; success consumes that version, so a stale read can never become a write.

WHAT KALDRIX DOES

Four powers. One desk.



Limits that hold

A person signs how much, to whom, how fast and until when. A tricked agent cannot go past it, and a person approves the big ones.

PER PAYMENT, TOTAL AND SPEED LIMITS



Data on a leash

An agent may read these tables, these columns or these rows, for this purpose, for this long. Anything else is refused before the database sees it.

PURPOSE AND TIME BOUND DATA ACCESS



Permissions that can only shrink

Signed, limited, expiring and cancellable on the spot. A helper can never hold more than the agent that made it.

ATTENUATING CAPABILITY CHAINS · ED25519



A record nobody can quietly rewrite

Every decision is written down with its reason, linked to the last so any edit shows.

SHA-256 HASH CHAINED AUDIT LOG

INSIDE A DECISION

Eight checks, one answer, in a fraction of a millisecond.



IN PLAIN WORDS

Every request is checked eight ways, then answered with a yes, a no and the reason, or a call for a human.

UNDER THE HOOD

A full decision takes about 131 µs at the median (measured on one laptop). The same checks guard a payment and a database statement.

KALDRIX GUARDS DATA TOO

An agent allowed to read tickets cannot read payroll.

THE SIGNED PERMISSION

PURPOSE
customer support

TIME
60 seconds, then it ends

TABLES
support tickets: read

COLUMNS
customers: id, name, plan

ROWS
ticket 7: change its status

CHECKED OUTSIDE THE AGENT, EVERY STATEMENT

Read the open tickets		ALLOWED
Read the payroll table	not a table it was given	REFUSED
Join tickets to payroll	every table a query touches counts	REFUSED
Close every ticket at once	it may read, and change one ticket	REFUSED
Drop the tickets table	schema changes are never allowed	REFUSED
The first query, 61 seconds later	the time ended while it was still running	REFUSED

IN PLAIN WORDS

The same signed permission that limits a payment limits a database. A tricked agent asks for payroll and the database never hears about it.

UNDER THE HOOD

Statement parsed under four SQL dialects, needs worked out per table, proof bound to the exact text. One pretend database.

PERMISSIONS

Authority can be handed down. It can never grow.



Narrow, never widen

A copy can only hold less than the permission it came from.

Cancel one, cancel all beneath

Pull a permission and every copy handed down from it stops at once.

Signed and personal

Each permission names its holder, expires on its own, and cannot be edited.

IN PLAIN WORDS

Give a team a budget and every agent below it gets only a slice. Nobody can quietly give themselves more.

UNDER THE HOOD

Macaroon-style HMAC-SHA256 chains with attenuation-only caveats; revocation propagates to every delegate and survives restart.

SHARED MONEY

Money is held before it is spent.

ALL-OR-NOTHING ACROSS THE WHOLE CHAIN OF DELEGATION



A team allowance of $50,000

1 Reserve

The moment an agent asks, the money is set aside. Nobody else can count it.

2 Capture

When the action succeeds, the hold becomes a real, final spend.

3 Release

If it fails or times out, the hold returns to the pool for others to use.

IN PLAIN WORDS

Like a hotel putting a hold on a card: the money is promised to one agent before the next one is served.

UNDER THE HOOD

Counters in the database (SQLite or PostgreSQL), reserved then kept or given back, tested with several connections at once. The earlier engine was also model checked.

AT SCALE

The same desk, from one agent to a whole network.



One agent

A single helper doing a single job.

- Limits on money ✓
- Limits on data ✓
- Shrinking permissions ✓
- Permanent record ✓



A team

Helpers sharing one budget.

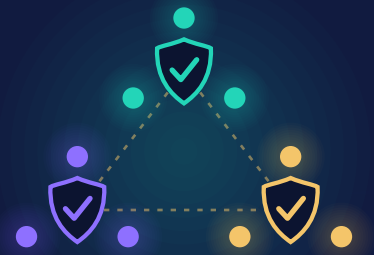
- Limits on money ✓
- Limits on data ✓
- Shrinking permissions ✓
- Permanent record ✓



A fleet

Departments of agents running whole processes.

- Limits on money ✓
- Limits on data ✓
- Shrinking permissions ✓
- Permanent record ✓



A network

Agents from different companies dealing with each other.

- Limits on money ✓
- Limits on data ✓
- Shrinking permissions ✓
- Permanent record ✓

IN PLAIN WORDS

Whether it is one helper or many, the same desk applies the same rules, and every function keeps working.

UNDER THE HOOD

Verification is stateless signed-token checking; shared state lives only in the spend ledger and the audit log. Tested under 16 and 24 simultaneous payments.

THE RECORD

A record nobody can quietly rewrite.



What is written

Who asked, what they asked for, the verdict and the reason, and the time. For every decision, allowed or refused.

How it is protected

Each record carries a fingerprint of the one before it. The log can be added to, never edited in place.

How you check

One call recomputes the whole chain. If anything was touched, it says exactly where.

IN PLAIN WORDS

Every decision is written down with its reason. If anyone changes the story afterwards, the change shows.

UNDER THE HOOD

Append-only log (the database refuses edits and deletes); SHA-256 hash chain; one call recomputes the chain and returns the head to store elsewhere.

SPEED

Fast enough to sit in front of every single action.



131 µs
median time for a full decision: chain, holder, proof, replay and scope

≈ 7,000
full decisions per second from one thread of one laptop

0
wrong answers across 20,000 timed decisions in every test row

IN PLAIN WORDS

The check takes a tiny fraction of a blink, so putting it in front of every action costs nothing you would notice.

UNDER THE HOOD

Ed25519 strict verification dominates the cost (about 65 µs per delegation block). Measured 17 Oct 2026, release build, one Windows laptop, one caller at a time.

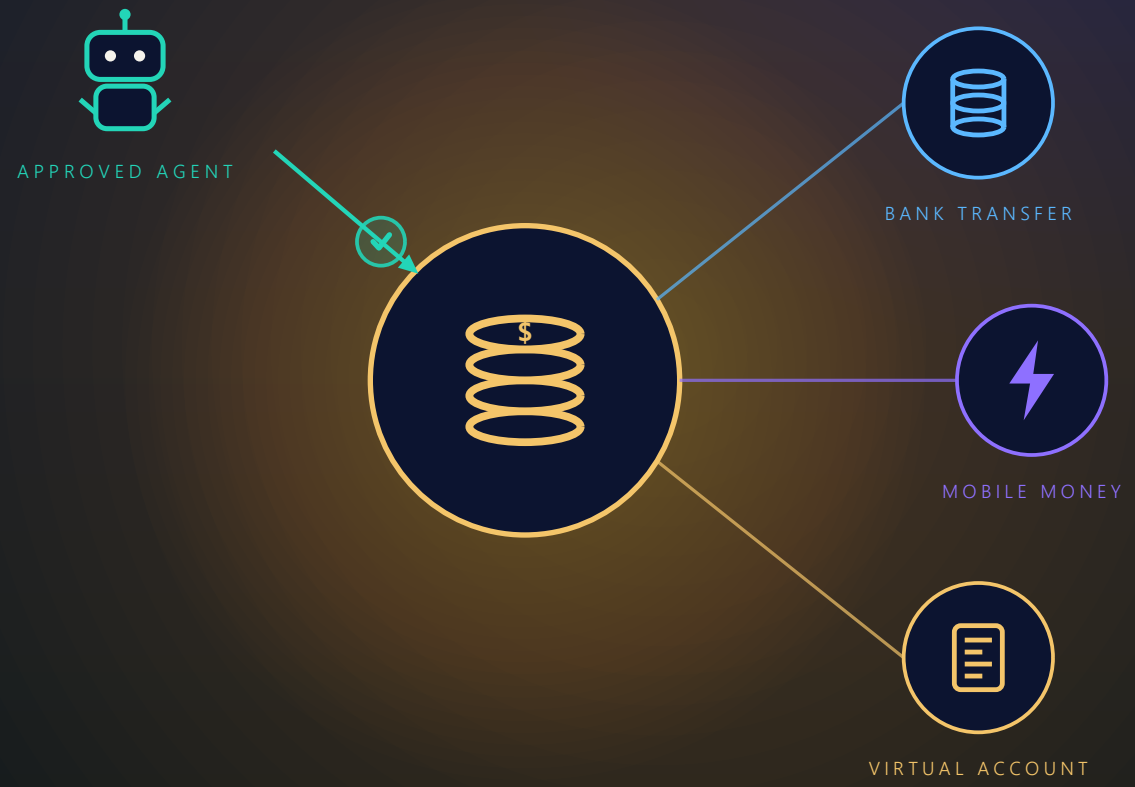
PART TWO · CONDITIONAL ESCROW AND EXECUTION CLEARINGHOUSE

Zentrix

Where AI agents move money: safely, locally, globally.

Kaldrix decides what an agent may do. Zentrix carries it out on the payment rails people actually use.

BUILT FOR GLOBAL USE | NIGERIA FIRST | PLUGS INTO ANY AUTHORITY



ONE GATEWAY, MANY LOCAL RAILS

WHAT ZENTRIX DOES

Four promises for every payment.



Pays on local rails

Bank transfer, mobile money and virtual accounts, country by country. A new market is a new adapter, not a rewrite.

CORRIDOR = COUNTRY · CURRENCY · RAIL



Pays exactly what was approved

The approval names the amount, the recipient and the purpose. Change any of it and the payment stops.

SIGNED CONSTRAINTS · TRANSACTION DIGEST



Never pays twice

Retry as often as you like. The same request always returns the same payment, and one approval pays once.

IDEMPOTENCY · ATOMIC LIMITS · INTEGER MINOR UNITS



Always explainable

Every payment has a status you can trust, a receipt, and a trail back to who approved it.

STATE MACHINE · RECEIPTS · HASH-CHAINED AUDIT

IN PLAIN WORDS

Money moves only when it is allowed, only as far as it was allowed, and only once.

UNDER THE HOOD

Amounts are integers in each currency's minor unit; limits are checked atomically with the state change; callbacks are idempotent; receipts come from stored data.

CONDITIONAL ESCROW

Money is held until the task is proven.

1 Lock

Funds are set aside for one exact task. The instructions are fingerprinted and the expected outputs are listed.

2 Prove

The agent signs a statement with its own key over the task and the fingerprints of what it produced.

3 Approve

If asked for, a named person says yes. The agent can never approve itself.

4 Release

Kaldrix is asked again, so a cancelled permission stops it. Then it pays, once.

WHAT STOPS IT

- A different file from the one the task required
- A proof signed by a key that is not the agent's
- A proof made for another escrow or another task
- A permission cancelled after the work was done
- The deadline passing: it lapses and goes back

THE VAULT IS A PRETEND HOLD

IN PLAIN WORDS

The agent has to show its work. Saying it is finished is not enough, and the money does not move on a promise.

UNDER THE HOOD

Ed25519 proof over escrow, task hash, time and output hashes; release by compare and set so only one caller can win; Kaldrix asked again at release.

INSIDE A PAYMENT

Seven stages. Every one on the record.

1

Request received

The instruction arrives with its own idempotency key.

2

Request validated

Well formed, amounts in whole minor units, key claimed.

3

Authorization checked

Kaldrix, or your own authority, must say yes to this exact payment.

THE SECURITY GATE

4

Payment rules checked

Limits, approved recipients and the corridor are enforced.

5

Sent to the provider

Only now does a provider see it. Accepted is not settled.

6

Provider reports back

Success, failure or unknown, reported as it is.

7

Final status confirmed

Settlement is confirmed and a receipt is produced.

REFUSED AT ANY STAGE

It stops there, with a reason on the record. The payment provider never sees it.

IN PLAIN WORDS

A payment is a short, visible journey. At any point you can see where it is and why it stopped.

UNDER THE HOOD

Each stage is derived only from events persisted for that payment; a stage is shown as done only if an event says so.

EXACT-PAYMENT BINDING

Approved for this payment. Not for any other.

THE SIGNED APPROVAL

AMOUNT
$25,000.00

RECIPIENT
Ada O.

PURPOSE
Supplier payment

REFERENCE
inv-1042

SEALED WITH 7c1e…b94f

PAYMENT A · SAME AS APPROVED

$25,000.00 to Ada O. for supplier payment

Fingerprint 7c1e…b94f matches the seal

PAID, ONCE

PAYMENT B · ONE DETAIL CHANGED

$250,000.00 to Ada O. for supplier payment

Fingerprint e03a…1d72 does not match the seal

REFUSED · authorization_binding

IN PLAIN WORDS

The approval is for one exact payment. A stolen or altered approval cannot be used to pay anything else.

UNDER THE HOOD

A SHA-256 digest of the payment plus signed amount, currency, recipient and purpose; a mismatch on any gets its own rejection code.

BRING ANY AUTHORITY

Zentrix trusts only the checks it can see were done.

Kaldrix

Signed, exact-payment decisions. Verified end to end.

Another platform

Plug in its adapter. It declares what it can check.

Your own service

Use the approval system you already run.

EACH PROVIDER DECLARES WHAT IT CAN CHECK

- ✓ Who signed it
- ✓ Has it expired
- ✓ Has it been revoked
- ✓ Is the amount allowed
- ✓ Is the currency allowed
- ✓ Is the recipient allowed
- ✓ Is the purpose allowed
- ✓ Is it this exact payment

A yes that skipped a check the account requires is treated as a no. A provider that cannot be reached, or gives no clear answer, is a no.

FAIL CLOSED

NEVER CLAIMS A CHECK IT DID NOT DO

IN PLAIN WORDS

Zentrix does not force one security system on you. It works with yours, and says plainly what each one checked.

UNDER THE HOOD

Declared capabilities and performed checks per provider; required checks per account; the Kaldrix adapter verifies Ed25519 signatures with public keys only.

BESIDE THE PROTOCOLS

Four doors into Zentrix. Each one is checked.

x402 facilitator

A web payment is valid only if it carries a Kaldrix permission. A well formed payment to the wrong address, over the limit, or with no permission is refused.

SIMULATED — No blockchain. Made up token. Wallet signature not checked.

AP2 mandate check

Checks that a closed payment satisfies the constraints of the open mandate: amounts, payees, budget, dates. An unknown constraint fails.

SIMULATED — Does not verify signatures. AP2 version 0.2 is pre release.

Card authorization sidecar

Before a card platform clears an agent's card payment, it asks whether it matches what the agent was approved for. Once, in time.

SIMULATED — No card network or issuer is connected.

Local rails: Nigeria, Kenya, Brazil

Quotes and receipts for NIP, Mpesa and PIX, with the destination checked and only the last four digits kept.

SIMULATED — Fixed example rates. Receipts are marked as simulated.

IN PLAIN WORDS

These are working code written against the public specifications. Every one runs as a simulation today, and none of the companies behind them is a partner.

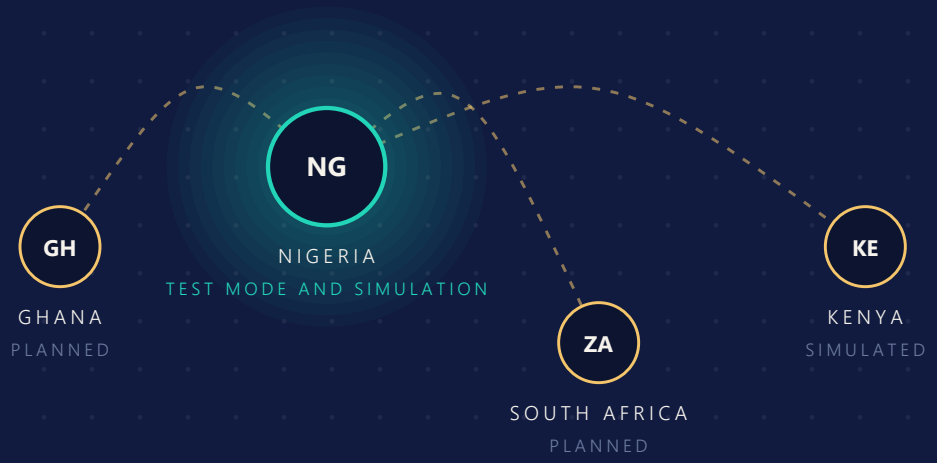
UNDER THE HOOD

x402 version 2 field names; AP2 version 0.2 constraints; compare and set stores; nothing is connected to a network, a bank or a card issuer.

GLOBAL BY DESIGN

Built for the world. Starting where it matters first: Nigeria.

NIGERIA FIRST



12 CURRENCIES CARRIED BY THE ENGINE, EACH WITH ITS OWN DECIMALS AND LIMITS

NGN · GHS · KES · ZAR · EGP · TZS · UGX · XOF · XAF · USD · EUR · GBP

HOW A MARKET IS ADDED

1. **Add the payment provider's adapter**
   One adapter per provider: the bank, the mobile-money operator, the local rail.
2. **Declare its corridor**
   A corridor is a country, a currency and a rail.
3. **Set limits and switch it on**
   Per currency, per account. Nothing is on by default.

THE PAYMENT ENGINE ITSELF NEVER CHANGES.

IN PLAIN WORDS

A new country is a new adapter and a setting, not a rebuild. The safety rules travel with every corridor.

UNDER THE HOOD

Country · currency · rail corridors; per-currency limits; routing by declared corridor. A corridor with no provider is refused, never rerouted.

SAFE BY CONSTRUCTION

Six habits that keep money from going wrong.



Whole-number money

Amounts are whole numbers of the smallest unit, so rounding errors cannot happen.



Same request, same payment

A retry returns the original. Reusing a key for a different payment is refused.



Limits that hold under pressure

Single-payment and daily limits are enforced atomically, tested with 24 payments at once.



Honest about the unknown

If a provider does not answer, the payment is marked unknown: never guessed paid or failed.



Reconciliation

Every recorded payment can be compared with the provider's own statement, and any difference shown.



Receipts from stored facts

A receipt is generated from what was recorded, never from what a caller claims.

IN PLAIN WORDS

Money software fails in small, boring ways. Each of these habits closes one of them.

UNDER THE HOOD

Integer minor units; unique idempotency keys; atomic limit checks; a validated state machine with an unknown-outcome state; receipts from stored rows.

THE OPPORTUNITY

Agents are about to move money. Who is in charge?

TODAY

Agents act



They book, buy, update and send on their own.

NEXT

Agents pay



The same agents settle invoices, run payouts and move funds.

ALWAYS

Someone must answer



For every payment, a person will be asked who allowed it.

THE QUESTION EVERY BANK, REGULATOR AND FINANCE TEAM WILL ASK

Who approved this payment, for how much, to whom, and can you prove it?

IN PLAIN WORDS

The more agents handle money, the more this question matters. Kaldrix and Zentrix answer it by design, on every payment.

UNDER THE HOOD

Signed decision bound to the exact payment, a verified chain of authority, and a tamper-evident record: the evidence exists before anyone asks for it.

WHO IT IS FOR

Anyone who lets software spend money.



Fintechs and payment apps

Let customers' AI assistants pay bills and suppliers, inside limits the customer set.

PER-CUSTOMER LIMITS



Payroll and payouts

Agents run payout batches under a budget. Nothing leaves that was not approved.

BUDGETS · APPROVALS



Marketplaces and platforms

Pay sellers, drivers and creators automatically, with a full trail for every payment.

AUTOMATIC PAYOUTS · FULL TRAIL



Enterprises running agent fleets

Give every agent its own budget, its own data access and audit trail, controlled from one desk.

FLEET WIDE CONTROL

IN PLAIN WORDS

If an AI can spend it, someone has to set the limits, check each payment and keep the proof. That is the job.

UNDER THE HOOD

Delegated budgets, exact-payment approval and an audit trail, exposed as an HTTP API with a TypeScript SDK; the authority is pluggable.

THE CONTROL ROOM

See every decision, as it happens.



LAYOUT OF THE CONTROL ROOM · FIGURES SHOWN ARE ILLUSTRATIVE

Two guided labs

Seven security scenarios and seven payment scenarios, run against the live system with the real result shown.

Live stream

Watch decisions arrive as they are made.

Audit and verification

Search every decision and recompute the chain on demand.

Security and authorization

Which authority is configured, and exactly what it can verify.

IN PLAIN WORDS

Everything the system decides is visible in one place, in plain language, with the reason beside it.

UNDER THE HOOD

React single-page app served by the platform; live stream over server-sent events with single-use tickets; every figure is counted from persisted events.

SEE FOR YOURSELF

Do not take our word for it. Open it.



SCAN, OR VISIT

kaldrix-proof.vercel.app

A RECORDED RUN, STEP BY STEP

Thirty six situations in which software agents handle money and data, each a real run of the real system. Step through and watch what Kaldrix decides.

AND FOUR WAYS TO TEST IT YOURSELF

A live test service	Make a private account, give an assistant a limited key, watch it pay and be refused. Pretend bank.
Break the Agent	A public challenge: try to make a signed limit fail.
The public connector	Install it in Claude, Cursor, Copilot or Codex in about five minutes.
Database access	Give an agent a data permission for 60 seconds and watch it read tickets and be refused payroll.

IN PLAIN WORDS

Open the link on any phone or laptop and follow the system making its decisions, one step at a time.

UNDER THE HOOD

The proof page uses a fixed clock and randomness, so re-running gives the identical recording. The live service is a pretend bank.

HONEST STATUS

What is real, what is simulated, what is not built.

BUILT AND TESTED

- ✓ Signed permissions that only narrow
- ✓ Limits per payment, in total and by speed
- ✓ A person approves the big ones
- ✓ A permission being probed freezes itself
- ✓ Database access by table, column and row
- ✓ Purpose and time limits on every request
- ✓ A tamper evident record
- ✓ A connector for Claude, Cursor, Copilot, Codex

WORKS AS A SIMULATION

- ✓ The bank (and Paystack's test mode)
- ✓ The escrow vault
- ✓ The x402 facilitator
- ✓ The AP2 constraint check
- ✓ The card authorization sidecar
- ✓ Nigeria, Kenya and Brazil payouts
- ✓ The one demo database

NOT BUILT, NOT CLAIMED

- ⊗ An independent human security review
- ⊗ Real money (needs a licensed partner)
- ⊗ A reviewed real database connector in use
- ⊗ A real escrow vault and real rails
- ⊗ Customers and revenue
- ⊗ Any partnership with a named company

IN PLAIN WORDS

We would rather say what is missing than have you find it. Everything marked as a simulation is labelled on the product itself.

UNDER THE HOOD

Open risks are ranked in docs/security/open-risks.md. Tests: about 1,200 Rust tests pass and the linter is clean; the hosted service has one pretend database; a read only PostgreSQL connector exists but is off and unreviewed.

BUILT, MEASURED, RUNNING

What exists today.

1,192
automated Rust tests passing across the platform, with the linter clean

17
required payment-safety tests, all passing

30
recorded situations, plus a live test service you can break

131 µs
median time for a full authorization decision

12
currencies carried, each exact to its smallest unit

3 of 3
high severity findings from a separate AI assisted review, fixed and re tested. No independent human review yet

WHAT COMES NEXT

An independent review
A human security practitioner

First design partners
Real workflows, in test mode first

One live payment partner
Small amounts, after the review

IN PLAIN WORDS

The foundation is built and tested, for money and for data. What remains is independent review, real users and a licensed partner to carry real money.

UNDER THE HOOD

Rust workspace of 19 crates, SQLite or PostgreSQL, React control room, TypeScript and Python SDKs, a public connector. Hosted as a test service.

Every agent.
Every action.
Every payment.

Decided by Kaldrix. Moved by Zentrix. Remembered for good.



CHECKED PROTECTED REMEMBERED

SEE IT RUN · kaldrix-proof.vercel.app